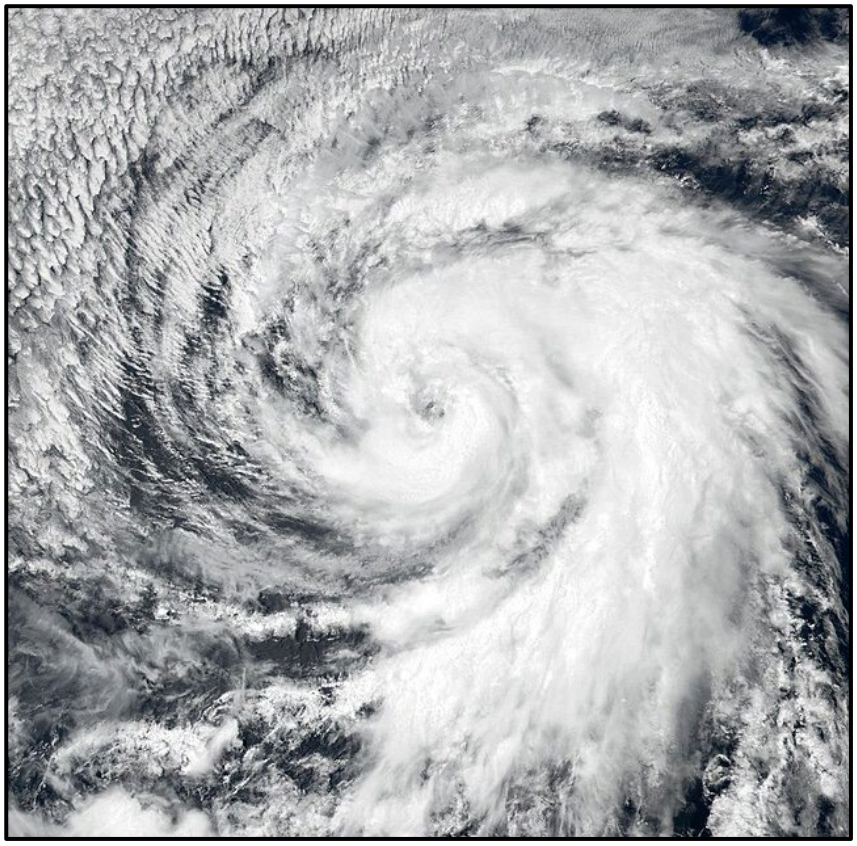
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE GIL

(EP072025)

31 July – 3 August 2025

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National Hurricane Center
15 September 2025



VISIBLESATELLITE IMAGE FROM NOAA'S SUOMI NPP OF GIL AT 2100 UTC 1 AUGUST. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Gil was Category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that remained over the open waters of the eastern North Pacific Ocean.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Gil.

Hurricane Gil

31 JULY – 3 AUGUST 2025

BEST TRACK

The "best track[2]" positions and intensities are listed in Table 1. The best track chart of Gil's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no land-based or ship reports of tropical-storm-force winds associated with Gil.

Origin

Gil formed around 0000 UTC 31 July about 650 n mi south-southwest of the southern tip of the Baja California peninsula. Gil appears to have formed from a combination of a tropical wave that crossed Central America on 26 and 27 July and a disturbance in the monsoon trough.

Peak Intensity and Minimum Pressure

Gil's peak intensity of 65 kt from 0000 to 1200 UTC 2 August is based on T4.0/65 kt Dvorak classifications from both TAFB/SAB and SATCON during that time period. The minimum central pressure of 991 mb is primarily based on the Knaff-Zehr-Courtney pressure wind relationship.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Gil.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), and Defense Meteorological Satellite Program (DMSP) satellites, among others, were also useful in constructing the best track of Gil.

FORECAST AND WARNING VERIFICATION

Table 2 lists the number of hours prior to formation corresponding to the first issuance of an NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The formation of Gil was generally well anticipated. The precursor disturbance was first introduced in the 7-day TWO with a low probability of genesis 138 h before development. This likelihood was subsequently increased to the medium category 120 h in advance, and to the high category 72 h prior to formation. Lead times for the 2-day outlooks demonstrated notable reliability. A low probability was first indicated 72 h before genesis, with increases to medium and high probabilities at 60 h and 48 h prior to development, respectively. Figure 4 shows composites of the 7-day TWO genesis areas for each likelihood category leading up to the formation of Gil. The anticipated formation region turned out to be too far east for the first few genesis predictions.

A verification of NHC official track and intensity forecasts for Gil are given in Table 3a and 4a, respectively. A homogeneous comparison of the official track and intensity errors with selected guidance models are given in Table 3b and 4b, respectively. The mean official track forecast errors for Gil were excellent, and well below the long-term averages. The only models that consistently outperformed NHC's track forecasts were EMXI and HCCA. The NHC mean official intensity errors for Gil were also quite low, and well below the 5-yr averages. Most of the hurricane regional models performed quite well for this system too.

There were no coastal watches or warnings issued for Gil.

Table 1. Best track for Hurricane Gil, 31 July–3 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
31 / 0000	12.5	113.3	1005	35	tropical storm
31 / 0600	12.7	114.5	1003	40	"
31 / 1200	12.9	115.6	1001	45	"
31 / 1800	13.3	116.7	999	50	"
01 / 0000	13.8	117.9	996	55	"
01 / 0600	14.3	119.3	996	55	"
01 / 1200	14.9	120.9	995	55	"
01 / 1800	15.5	122.6	993	60	"
02 / 0000	16.2	124.3	991	65	hurricane
02 / 0600	17.0	126.0	991	65	"
02 / 1200	17.7	127.8	992	65	"
02 / 1800	18.4	129.5	995	60	tropical storm
03 / 0000	19.2	131.2	998	55	"
03 / 0600	19.8	132.9	1001	50	"
03 / 1200	20.3	134.8	1003	45	low
03 / 1800	20.6	136.6	1005	40	"
04 / 0000	20.8	138.3	1006	35	"
04 / 0600	21.0	139.9	1007	30	"
04 / 1200	21.1	141.2	1008	30	"
04 / 1800	21.2	142.4	1008	30	"
05 / 0000	21.3	143.4	1009	25	"
05 / 0600					dissipated
02 / 0000	16.2	124.3	991	65	maximum wind and minimum pressure

Table 2. Number of hours in advance of formation of Gil associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	72	138
Medium (40%-60%)	60	120
High (>60%)	48	72

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Gil, 31 July–3 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**15.7**	**25.6**	**32.2**	**27.9**	**29.5**	**33.5**		
OCD5	36.0	103.1	189.7	280.2	357.2	380.9		
Forecasts	11	9	7	5	3	1		
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Gil, 31 July–3 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	15.4	29.1	36.4	32.5	40.0			
OCD5	37.1	115.7	220.9	315.2	427.5			
GFSI	28.0	57.2	62.1	70.4	49.6			
EMXI	**13.4**	**21.0**	**25.6**	**23.6**	**16.5**			
CMCI	25.0	44.0	71.9	88.9	99.4			
HWFI	30.8	57.5	70.3	79.8	107.9			
HMNI	26.7	51.6	57.3	59.6	**33.9**			
HFAI	28.1	53.9	65.7	74.0	52.2			
HFBI	31.1	57.6	67.5	80.8	84.4			
AEMI	27.4	55.8	66.3	68.7	73.6			
CTCI	32.8	66.9	78.0	90.7	101.8			
HCCA	**14.4**	**22.0**	**18.8**	**24.7**	**24.7**			
TVCE	22.0	41.2	46.3	47.6	40.0			
GDMI	17.6	32.1	36.9	41.5	45.2			
Forecasts	8	6	4	3	1			

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Gil, 31 July–3 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**2.7**	**6.7**	**6.4**	**7.0**	**6.7**	**5.0**		
OCD5	3.8	6.4	5.1	7.4	15.0	14.0		
Forecasts	11	9	7	5	3	1		
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Gil, 31 July–3 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.1	6.7	6.2	8.3	10.0			
OCD5	3.8	**6.5**	**3.8**	**7.7**	19.0			
HWFI	4.0	**5.2**	6.2	**2.3**	**1.0**			
HMNI	**2.8**	**6.2**	11.0	**7.0**	**7.0**			
HFAI	3.6	**4.5**	**3.2**	**1.7**	**0.0**			
HFBI	**3.0**	**6.2**	**6.0**	**2.3**	**0.0**			
CTCI	**1.8**	**4.7**	**5.0**	**4.0**	**3.0**			
HCCA	**2.6**	**3.2**	**3.2**	**4.0**	**4.0**			
IVCN	**1.9**	**4.0**	**2.5**	**1.3**	**3.0**			
GFSI	3.5	**3.8**	**4.2**	**3.0**	**3.0**			
EMXI	4.0	7.8	7.5	**5.7**	**4.0**			
GDMI	4.1	**4.7**	7.5	9.0	10.0			
DSHP	3.2	7.0	11.2	16.0	19.0			
LGEM	**2.8**	**3.8**	**1.0**	**3.0**	**5.0**			
Forecasts	8	6	4	3	1			

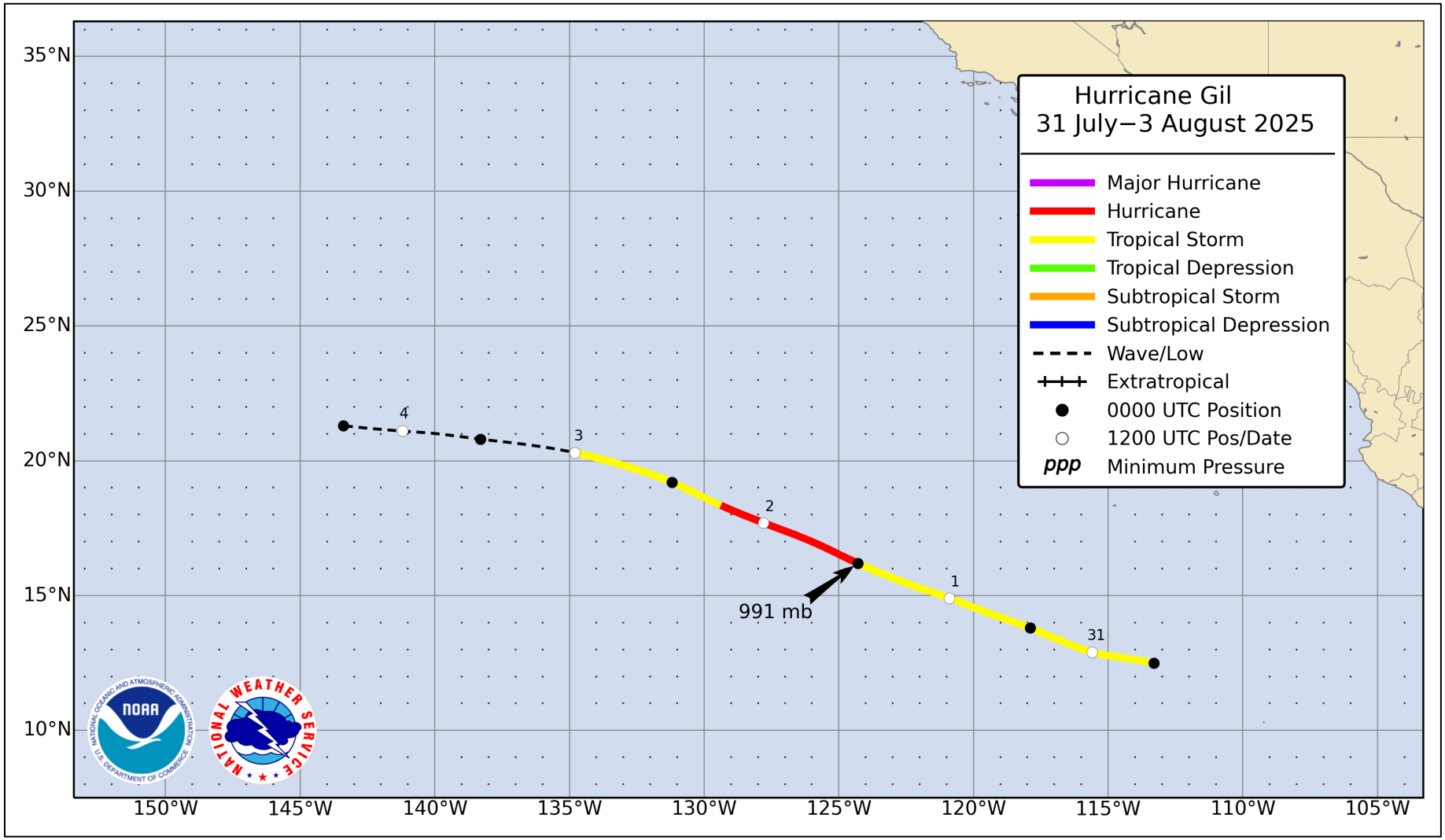


Figure 1. Best track positions for Hurricane Gil, 31 July–3 August 2023.

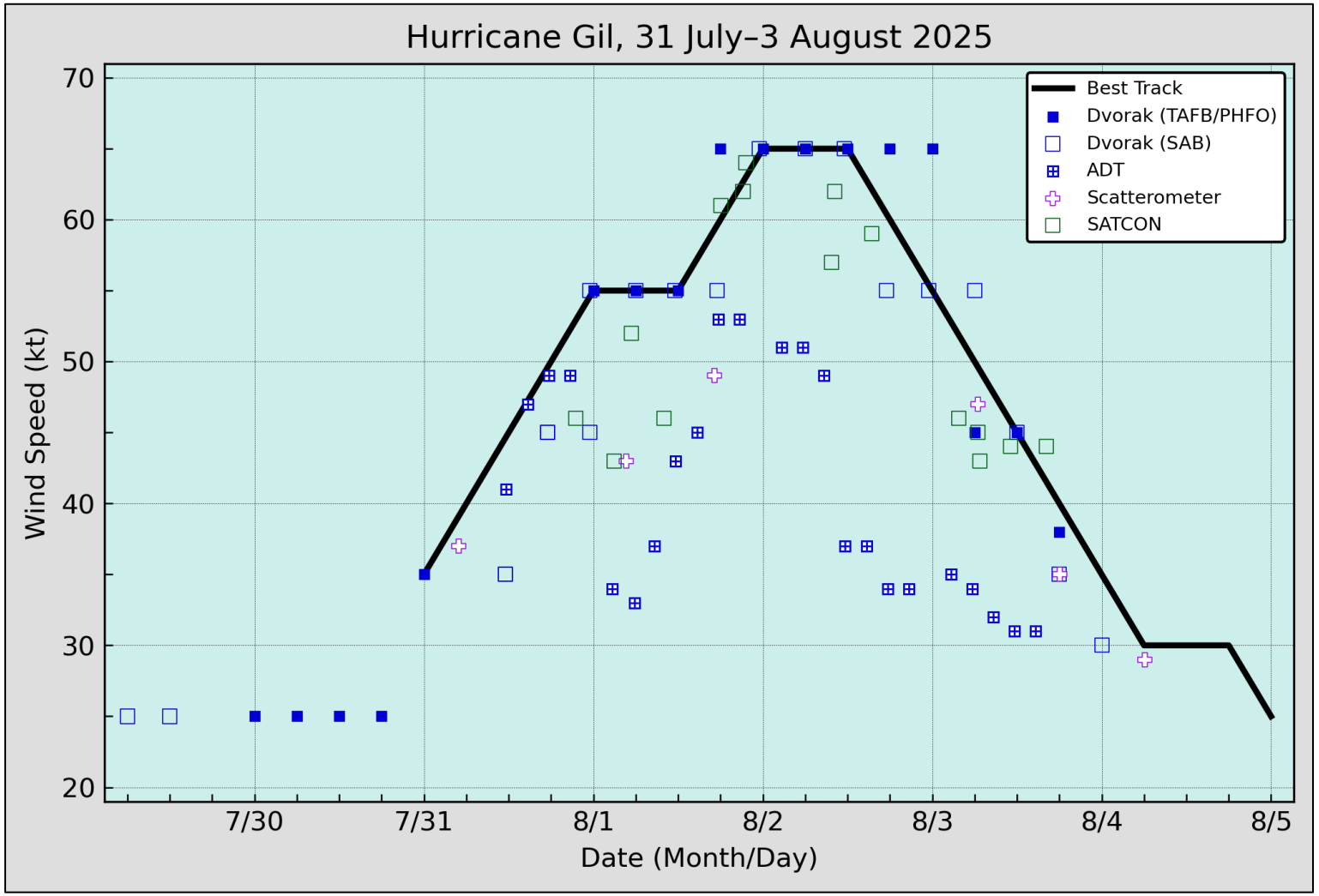


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Gil, 31 July–3 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

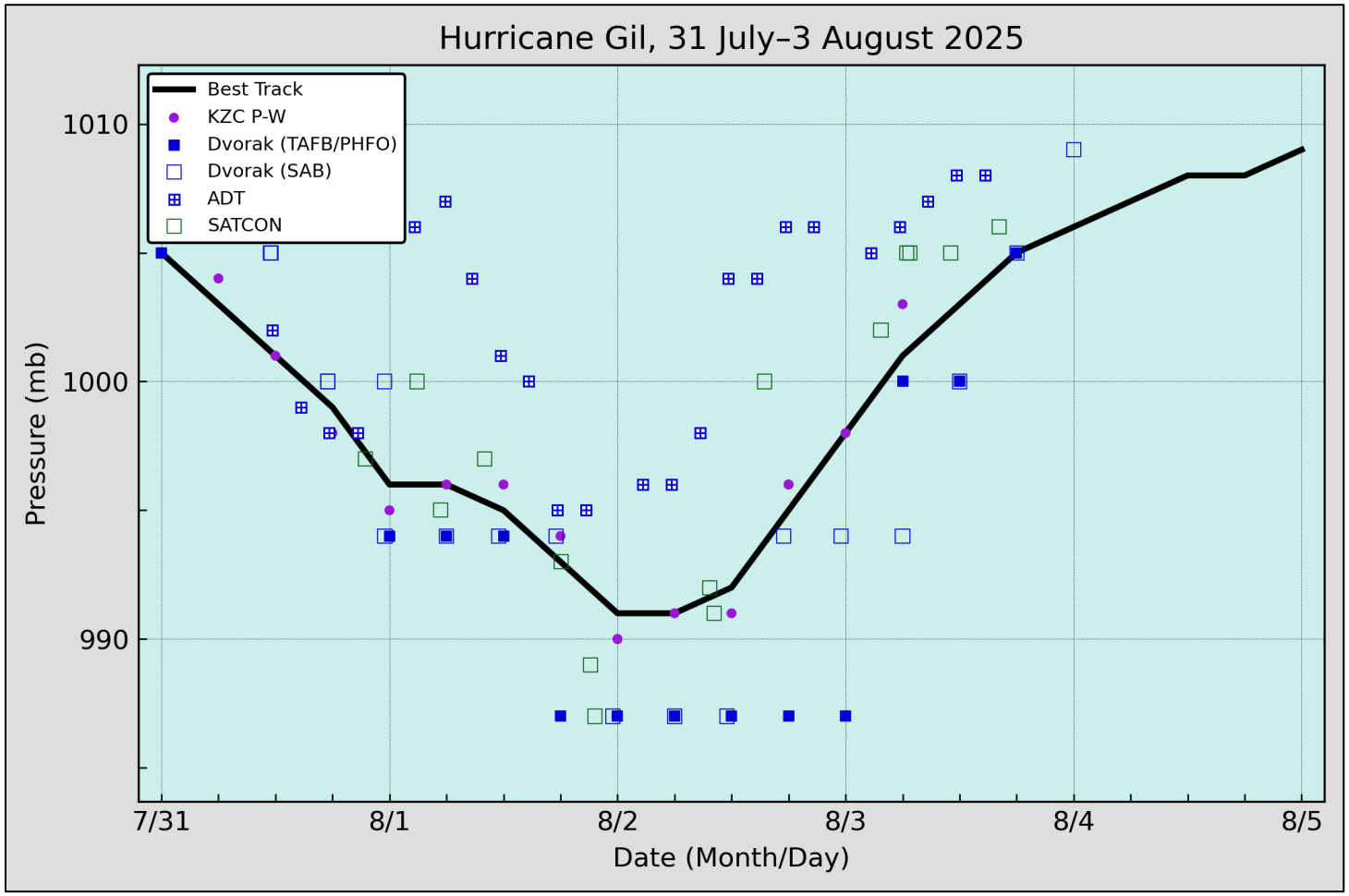


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Gil, 31 July–3 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

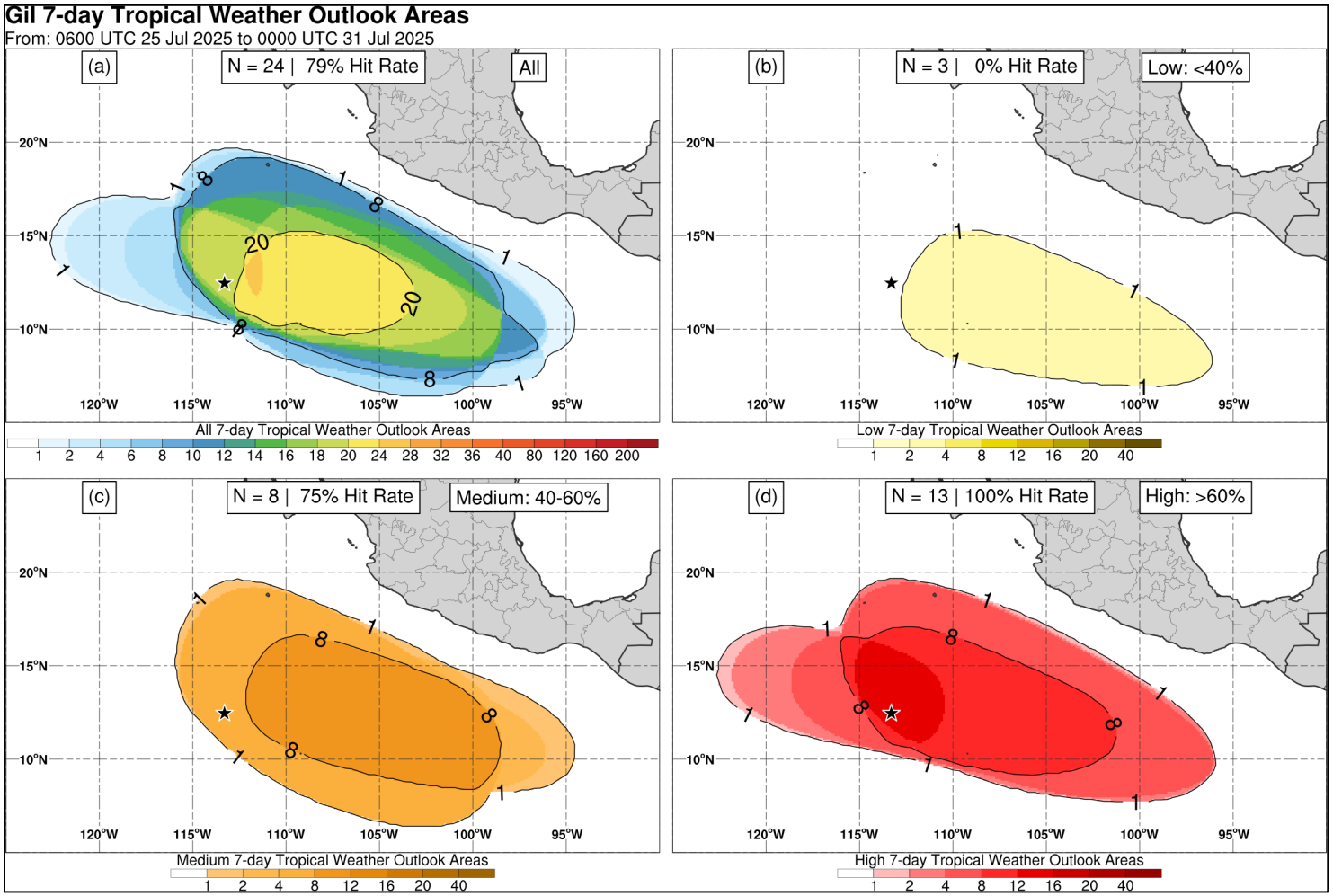


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Gil for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.